

Zhou Yinzhe

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https://stevenzhoooooooooooooooo.github.io/

EDUCATION

University of Macau

Bachelor of Science in Electrical and Computer Engineering

Macau, SAR

Aug. 2021 – May 2025

Honors: Dean's Honor List (All Semesters), Akrostar Academic Prize (2024/2025).

Relevant Coursework: Embedded Systems, Circuit Analysis, Digital Systems, Computer Programming, Network Fundamentals, Linear Algebra, Signals and Systems.

City University of Hong Kong

Exchange Student, Department of Electrical Engineering

Hong Kong, SAR

Jan. 2023 – Aug. 2023

Honors: Academic Excellence Outgoing Exchange Scholarship.

Relevant Coursework: Principles of Communications, SoC Design, Electric Machine and Drive, Bio-sensor and Device.

Johns Hopkins University

Master of Science in Electrical and Computer Engineering

Baltimore, MD, USA

Aug. 2025 – Present

GPA: 3.79

Relevant Coursework: FPGA Synthesis Lab, CAD Design of Digital VLSI Systems I, Advanced Topics In Microsystem Fabrication.

RESEARCH & PROJECT EXPERIENCE

University of Macau

Research Assistant

Macau, SAR

May 2022 – Jul. 2022

FPGA-based Hardware Time Triggered System for Biomedical Engineering

- Implemented sophisticated signal conditioning and filtering algorithms in the Quartus FPGA environment.
- Optimized hardware architecture and interfaces to ensure deterministic, time-triggered behavior for mission-critical biomedical systems.
- Tools: Quartus, Matlab.

City University of Hong Kong

Research Assistant

Hong Kong, SAR

May 2023 – Aug. 2023

Neural Decoding in Implantable Brain-Machine Interfaces with ANN and SNN

- Preprocessed raw monkey neural data to create a dataset labeled with movement coordinates, finger position, and neural spike signals.
- Designed and optimized the architecture of Artificial Neural Networks (ANN) and Spiking Neural Networks (SNN).
- Tools: Python (PyCharm), Matlab.

Microsoft Research Asia (MSRA)

Research Intern

Beijing, China

Jun. 2024 – Sep. 2024

Finetuning Mixtral 8x7B Language Model with Biochemical Molecular Data

- Extracted biochemical molecular information from the PubChem database to construct training datasets and performed rigorous data preprocessing.
- Retrieved molecular SMILES using KNN and reconstructed the dataset by incorporating similar molecules to enhance the model's In-Context Learning capabilities.
- Optimized the architecture and hyperparameters of the Mixtral 8x7B model; designed a new training/testing approach that increased test accuracy from **67% to 77%**.
- Tools: Python, Git, Visual Studio Code.

University of Macau

Final Year Project - Faculty Dean's Honors List

Macau, SAR

Sep. 2024 – May 2025

Singular Spectrum Analysis (SSA) for Biosignal Denoising and Feature Extraction

- Designed a two-level SSA-based strategy combining component correlation and waveform reconstruction to accurately localize PQRS features in ECG signals.
- Implemented an SSA method for EEG artifact removal and rhythm extraction, successfully isolating delta, theta, alpha, and beta bands while preserving neural activity.
- Validated approach on synthetic and real-world datasets, demonstrating effectiveness in capturing physiological patterns (e.g., alpha rhythm differences in eyes-open vs. eyes-closed states).
- Honors: Faculty of Science and Technology Final Year Project Dean's Honors List.
- Tools: Python, NumPy, SciPy, Matplotlib, NeuroKit2.

HONORS & AWARDS

Academic & Professional

- **Dean's Honor List:** Awarded for all semesters at University of Macau.
- **Akrostar Academic Prize:** Academic Year 2024/2025 (500 USD).
- **Academic Excellence Outgoing Exchange Scholarship:** 1,245 USD.
- **1st Runner-Up:** IEEE Student Branch Project Competition 2025.
- **Top 10%:** Huawei ICT Competition (Preliminary Round), Macau (2021-2022).

Athletics (Fencing - Men's Foil)

- **1st Place (1st of 34):** Savage Fencing Club Oktoberfecht Senior Mixed Foil, USA (Oct 2025). **US Fencing Rating: C25.**
- **1st Place:** 28th National University Fencing Championships (Group Tournament).
- **2nd Place:** 27th National University Fencing Championships (Group Tournament).
- **3rd Place:** Macau University Fencing Championship (2021-2022 Season).
- **Top 16:** Macau Fencing Ranking Competition (2021).

TECHNICAL SKILLS

Programming: Python, MATLAB, C/C++, Verilog, VHDL.

Tools: Vivado, Cadence, ROS (Robot Operating System), 3D CAD, Git.

Domain: LLM Fine-tuning, Signal Processing (EEG/ECG), Brain-Machine Interfaces, Embedded Systems, Robotic Learning, Digital Circuit Design.